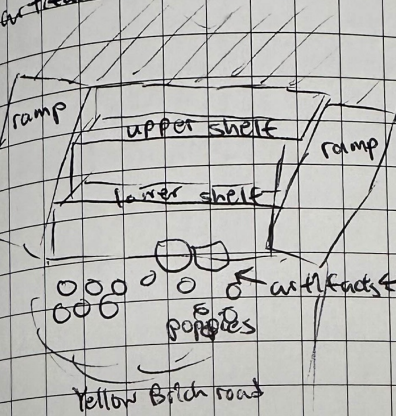


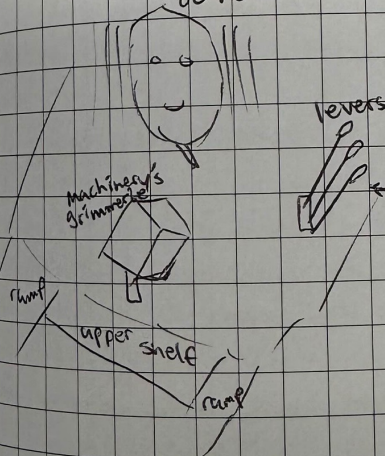
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Strategy 1: Stay on the ground and only collect poppies and artifacts



The robot will only operate below the ramps

Strategy 2: Push levers only + collect book + try oz head multiplier



Unlike strategy 1, the robot will only operate in range of the oz heart

- These 3 are close to each other, so less distance travel

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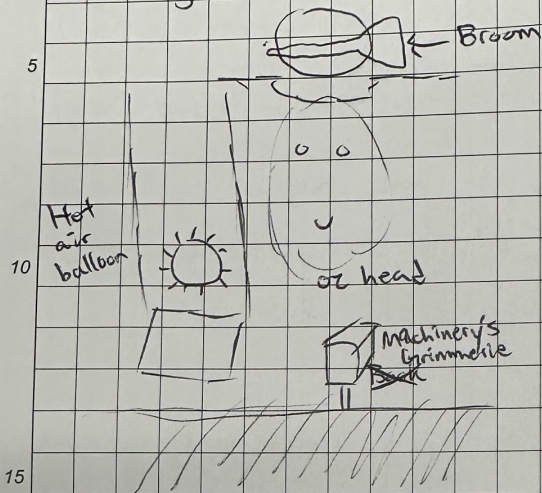
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Strategy 3: Aim for hot air balloon and the broom
 - go for the multiplier if enough time



- The robot will take the hot air balloon up to get to the Broom

Analysis

Strategy 1: Since the robot only operates on the base level, less time is used for travel.

- If it gets 4 artifacts on the ~~uppershell~~ uppershell and 15 poppies, then total point is $4 \cdot 35 + 15 = 155$ points

- Ignore bonus +5 just in case if there's not enough time to move the artifacts

- Most of the allotted time will be allotted for the artifacts, so 60s for artifacts and 30s for poppies

- Alternatively: Get 3 artifacts on uppershell, ~~15 poppies~~, and attempt the oz head multiplier. Points range between 126-210

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Strategy 2: Since each level awards a point per flip, the robot can specialize in flipping all 3 at once for triple the points. At some point near the end, the robot will attempt to collect the machinery's grimmerie and tilt the lever.

$$\text{Total points} = 3(90 - 5 - 10 - 20) \cdot (1.2 \text{ to } 2) + 23 \cdot (1.2 \text{ to } 2)$$

$\nwarrow$ # of levers $\uparrow$ total time $\uparrow$ time to get to the lever $\rightarrow$ mistiming of the lever $\uparrow$ multiplier $\uparrow$ machinery's grimmerie

- You can only flip the lever once per second, which allows us to simplify the calculation

- If we get the lever and the book but not multiplier, then total points = 188

- With 1.2x multiplier, total points = 225.6

- With 2x multiplier, total points = 376

Strategy 3: Use the hot air balloon to get to the Broom

- Similar to the previous strategy, this strategy reduces the distance traveled

$$\text{Total points} = 20 \cdot (1 \text{ to } 1.4) + 150 = 170 - 178$$

$\uparrow$ 4th zone $\uparrow$ multiplier $\uparrow$ Broom

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Pugh Chart

1 = excellent, 2 = good, 3 = decent

	Strategy 1	Strategy 2	Strategy 3
5 Points	2	1	2
Tipping	3	2	1
Stalling	1	3	3
10 Distance traveled	2	1	1
Ease of manufacturing	2	1	3
15 (base off of how the concepts I thought of in the moment)			
Total	10	8	10

- I choose strategy 2 since it has a lower score on the chart

Core requirements

- Robot must be able to push the lever since prim, and we don't want tipping

- The robot should move at least 2m/s to get to the destination at the right time and help push the levers

- Robot should drop off and reach objects at least a foot above ground.

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